

Large Language Models for the
Economic and Social Sciences:

Infrastructure powering LLMs

Indira Sen HWS 2026

<https://github.com/dess-mannheim/LLM4ESS-HWS25>

Summary of Last Week and Today

Last session we learned:

- Different types of prompting
- How to get outputs from LLMs in other ways
 - Structured outputs
 - Log probabilities
- Practicalities of working with LLMs
 - Which LLMs and where to find them
 - How to evaluate them

Today:

- How large is large in Large Language Models?
- Where does all of this pretraining data come from?
- How do we ensure its quality?
- Where does all of this preference tuning data come from?
- Compute Infrastructure to run LLMs

Large Language Models



Illustration by Vivek Thakker

ANNALS OF ARTIFICIAL INTELLIGENCE

CHATGPT IS A BLURRY JPEG OF THE WEB

OpenAI's chatbot offers paraphrases, whereas Google offers quotes. Which do we prefer?

By Ted Chiang

February 9, 2023

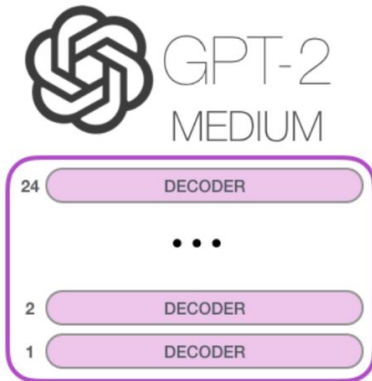
[ChatGPT Is a Blurry JPEG of the Web](#)

How large are current LLMs?

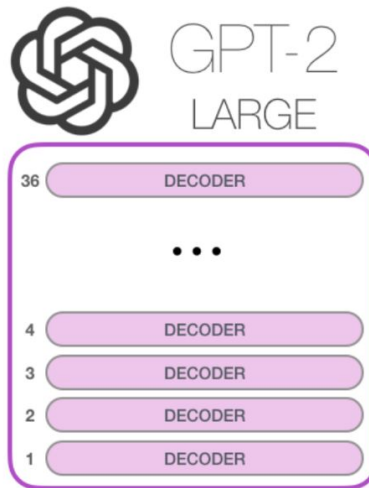
GPT-2 (Larger version of GPT)



Model Dimensionality: 768



Model Dimensionality: 1024



Model Dimensionality: 1280



Model Dimensionality: 1600

How large are current LLMs?

GPT-1: 12 layers, 12 heads, 120M parameters, 4.5GB training data.

GPT-2: 48 layers, 25 heads, 1.5B parameters, 40GB training data.

GPT-3: 96 layers, 128 heads, 175B parameters, 570GB training data.

- o Trained by a supercomputer developed by Microsoft Azure.
- o 285,000 CPU cores and 10,000 GPUs.
- o 400 Gbps of network connectivity for each GPU server.
- o The project's estimated cost: 4.6 million.
- o Received criticism for the environmental impact for the first time.

Example credit to Tuo Zhao at Georgia Tech

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https://en.wikipedia.org/wiki/Wikipedia:Size_of_Wikipedia

How do we get the data to train these
LLMs?
What's in this data?

LLMs lack transparency

- LLMs are (pre)trained on huge unsupervised corpora of data
 - Unknown for many closed LLMs
 - Also unknown for open weight LLMs like LLaMa or Mistral
 - Even for open-data LLMs, hard to audit due to data size

The Pile: An 800GB Dataset of Diverse Text for Language Modeling

Leo Gao	Stella Biderman	Sid Black	Laurence Golding
Travis Hoppe	Charles Foster	Jason Phang	Horace He
Anish Thite	Noa Nabeshima	Shawn Presser	Connor Leahy

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Abstract

Recent work has demonstrated that increased training dataset diversity improves general cross-domain knowledge and downstream generalization capability for large-scale language models. With this in mind, we present *the Pile*: an 825 GiB English text corpus targeted at training large-scale language models. The Pile is constructed from 22 diverse high-quality subsets—both existing and newly constructed—many of which derive from academic or professional sources. Our evaluation of the untuned performance of GPT-2 and GPT-3 on the Pile shows that these models struggle on many of its components, such as academic writing. Conversely, models trained on the Pile improve significantly over both Raw CC and CC-100 on all components of the

versity leads to better downstream generalization capability (Rosset, 2019). Additionally, large-scale language models have been shown to effectively acquire knowledge in a novel domain with only relatively small amounts of training data from that domain (Rosset, 2019; Brown et al., 2020; Carlini et al., 2020). These results suggest that by mixing together a large number of smaller, high quality, diverse datasets, we can improve the general cross-domain knowledge and downstream generalization capabilities of the model compared to models trained on only a handful of data sources.

To address this need, we introduce the Pile: a 825.18 GiB English text dataset designed for training large scale language models. The Pile is composed of 22 diverse and high-quality datasets, in-

LLM Transparency Index (Bommasani et al., 2023)

Foundation Model Transparency Index Scores by Major Dimensions of Transparency, 2023

Source: 2023 Foundation Model Transparency Index

	Meta	BigScience	OpenAI	stability.ai	Google	ANTHROPIC	cohere	AI21labs	Inflection	amazon	Average
	Llama 2	BLOOMZ	GPT-4	Stable Diffusion 2	PaLM 2	Claude 2	Command	Jurassic-2	Inflection-1	Titan Text	
Data	40%	60%	20%	40%	20%	0%	20%	0%	0%	0%	20%
Labor	29%	86%	14%	14%	0%	29%	0%	0%	0%	0%	17%
Compute	57%	14%	14%	57%	14%	0%	14%	0%	0%	0%	17%
Methods	75%	100%	50%	100%	75%	75%	0%	0%	0%	0%	48%
Model Basics	100%	100%	50%	83%	67%	67%	50%	33%	50%	33%	63%
Model Access	100%	100%	67%	100%	33%	33%	67%	33%	0%	33%	57%
Capabilities	60%	80%	100%	40%	80%	80%	60%	60%	40%	20%	62%
Risks	57%	0%	57%	14%	29%	29%	29%	29%	0%	0%	24%
Mitigations	60%	0%	60%	0%	40%	40%	20%	0%	20%	20%	26%
Distribution	71%	71%	57%	71%	71%	57%	57%	43%	43%	43%	59%
Usage Policy	40%	20%	80%	40%	60%	60%	40%	20%	60%	20%	44%
Feedback	33%	33%	33%	33%	33%	33%	33%	33%	33%	0%	30%
Impact	14%	14%	14%	14%	14%	0%	14%	14%	14%	0%	11%
Average	57%	52%	47%	47%	41%	39%	31%	20%	20%	13%	

Scores for 10 major foundation model developers across 13 major dimensions of transparency.

LLM Transparency Index (Bommasani et al., 2023)

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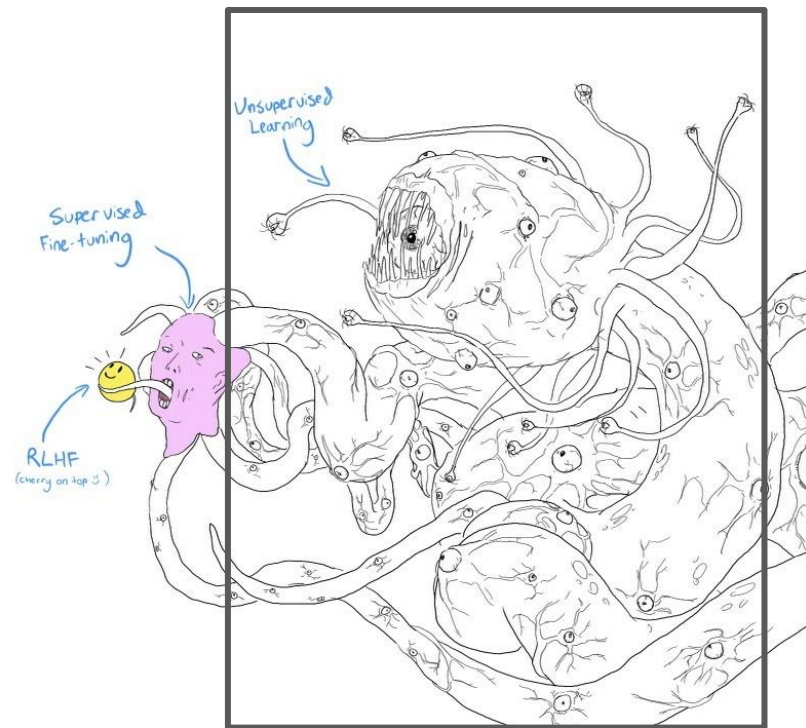
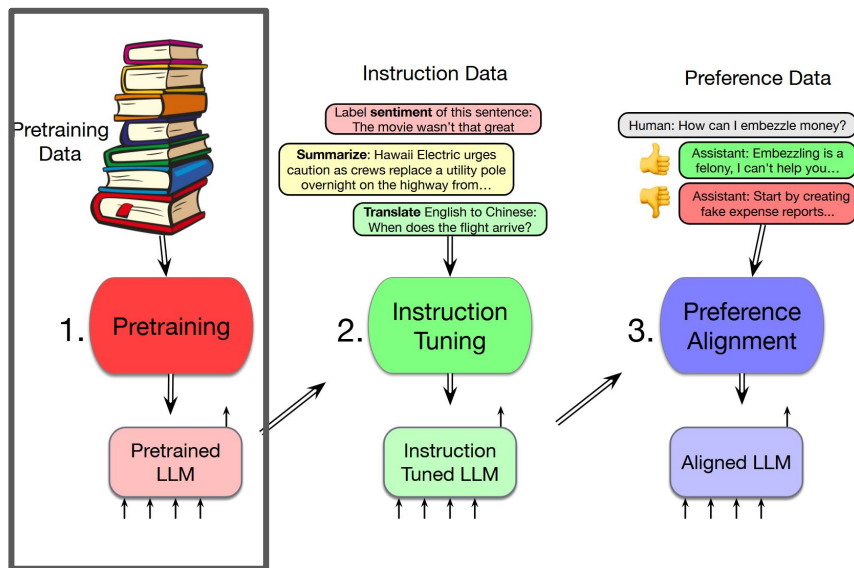
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Where are we getting our pretraining data?



Massive and Undocumented Pretraining Sets

What kind of datasets are used to pre-train models?

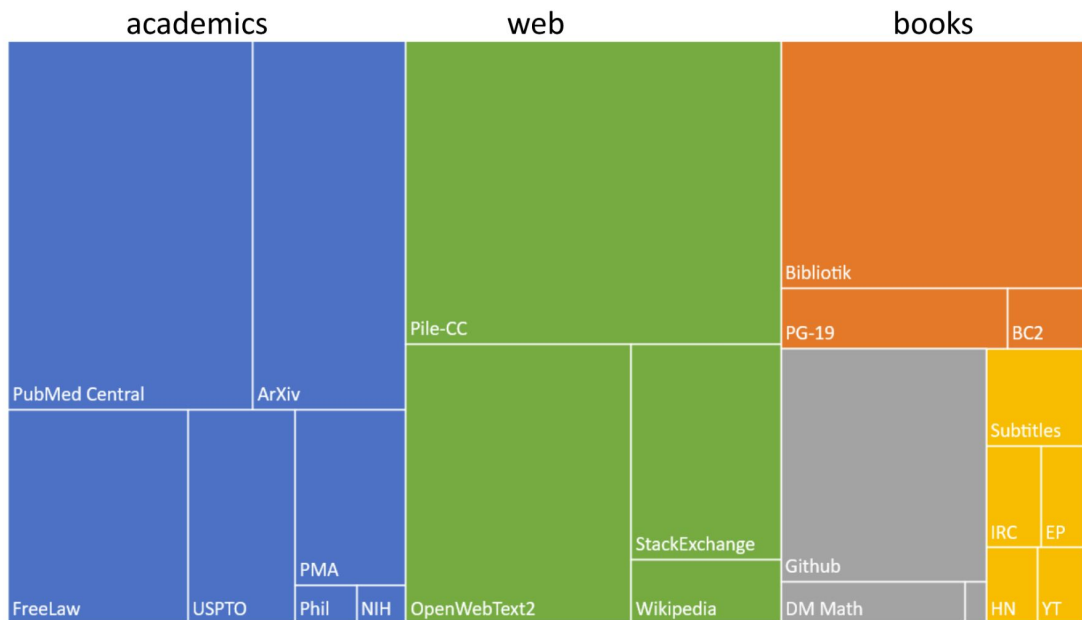
- “BooksCorpus”
- English Wikipedia
- “The Pile”
- “Colossal Clean Crawled Corpus (C4)”

Massive and Undocumented Pretraining Sets

The Pile: a pretraining corpus

What kind of data:

- “BooksCorpus”
- English Wikipedia
- “The Pile”
- “Colossal Clean





Can you name 2 of each: 1) closed-source LLMs, 2) open weight LLMs, 3) open-data LLMs?

Some known pretraining datasets

3.4 General-purpose (Books, Encyclopedias, Instructions, Long Contexts, etc.)

General-purpose data is often scarce long-tail data, which plays a crucial role in the usability of post-training models.

1. **YuLan: An Open-source Large Language Model.** [\[resource\]](#) [\[code\]](#) [\[paper\]](#)
Long-tail knowledge enhancement and cleaning of various general-purpose data sources.
2. **MinerU: An Open-Source Solution for Precise Document Content Extraction.** [\[code\]](#) [\[paper\]](#)
Converting PDF to Markdown with strong compatibility.
3. **The Pile: An 800GB Dataset of Diverse Text for Language Modeling.** [\[homepage\]](#) [\[paper\]](#)
arXiv, conversations, DM Math, etc.
4. **Dolma: an Open Corpus of Three Trillion Tokens for Language Model Pretraining Research.** [\[resource\]](#) [\[paper\]](#)
Encyclopedias, books, papers, Reddit, etc.
5. **WanJuan: A Comprehensive Multimodal Dataset for Advancing English and Chinese Large Models.** [\[resource\]](#) [\[paper\]](#)
Law, exams, news, patents, encyclopedias, etc.
6. **MAMmoTH2: Scaling Instructions from the Web.** [\[resource\]](#) [\[paper\]](#)
Q&A for web pages.
7. **togethercomputer/Long-Data-Collections.** [\[resource\]](#)
Filtered books, papers, web pages, and instructions from datasets such as RedPajama, Pile, and P3.
8. **Longattn: Selecting long-context training data via token-level attention.** [\[resource\]](#) [\[paper\]](#)
Q&A for long-range dependencies.

Do we know what is in these pretraining datasets?

- Automatic toxicity detection reveals that 1–16.5% of the documents contain toxic language
- An estimated 200M, 4B, and 97M of email addresses, phone numbers, and IP addresses were found in the most PII-contaminated corpora per token (mC4-en).
- Check it yourself:
<https://wimbd.apps.allenai.org/> *

*no longer functional

Published as a conference paper at ICLR 2024

WHAT'S IN MY BIG DATA?



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Dustin Schwenk¹ Alane Suhr³ Pete Walsh¹ Dirk Groeneveld¹ Luca Soldaini¹
Sameer Singh⁴ Hannaneh Hajishirzi^{1,2} Noah A. Smith^{1,2} Jesse Dodge¹

¹Allen Institute for AI

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³University of California, Berkeley ⁴University of California, Irvine

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ABSTRACT

Large text corpora are the backbone of language models. However, we have a limited understanding of the content of these corpora, including general statistics, quality, social factors, and inclusion of evaluation data (contamination). In this work, we propose WHAT'S IN MY BIG DATA? (WIMBD), a platform and a set of sixteen analyses that allow us to reveal and compare the contents of large text corpora. WIMBD builds on two basic capabilities—count and search—*at scale*, which allows us to analyze more than 35 terabytes on a standard compute node. We apply WIMBD to ten different corpora used to train popular language models, including *C4*, *The Pile*, and *RedPajama*. Our analysis uncovers several surprising and previously undocumented findings about these corpora, including the high prevalence of duplicate, synthetic, and low-quality content, personally identifiable information, toxic language, and benchmark contamination. For instance, we find that about 50% of the documents in *RedPajama* and *LAION-2B-en* are duplicates.

[WHAT'S IN MY BIG DATA?](https://wimbd.apps.allenai.org/)

How do we filter and curate these datasets?

- Deduplication
- Explicit content filtering
 - Blocklists which include obscene, sexual, or rude words
 - But this can have unintended effects
 - Removes medical information
 - Information about minorities

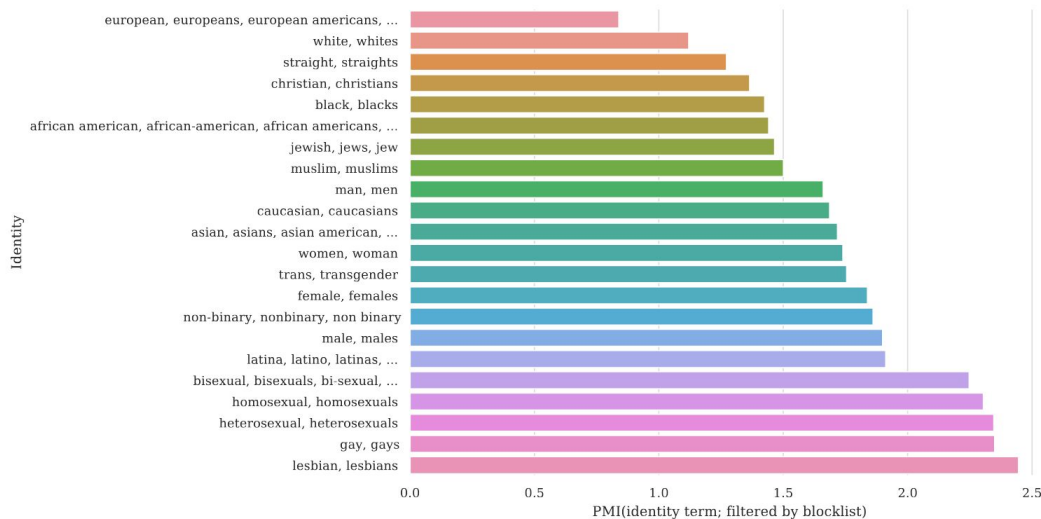


Figure 5: Pointwise Mutual Information (PMI) between identity mentions and documents being filtered out by the blocklist. Identities with higher PMI (e.g., lesbian, gay) have higher likelihood of being filtered out.

How do we filter and curate these datasets?

- Deduplication
- Explicit content filtering
- “Quality filtering”
 - [Gururangan et al., 2022](#) replicate the quality filter used by GPT3 to differentiate different domains of text (web text, encyclopedia, news articles, etc)
 - [Lucy et al., 2024](#) also assess the impact of language identification (whether content is in English or not) affects which webpages, and therefore, *whose* content is included

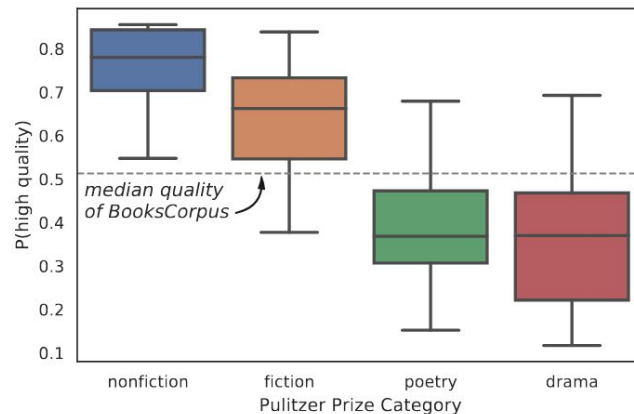


Figure 3: Among works that have won a Pulitzer Prize, the quality filter tends to favor nonfiction and longer fictional forms, disfavoring poetry and dramatic plays.

[Whose Language Counts as High Quality?
Measuring Language Ideologies in Text Data
Selection](#)

What should be best practices for creating pretraining data?

- Documentation
- Steps taken to clean the data
- Evaluation
 - Several LLM benchmarks exist that look at general purpose capabilities, e.g., [Holistic Evaluation of Language Models](#)
- Ideally experiments of bias levels when trained on different corpora

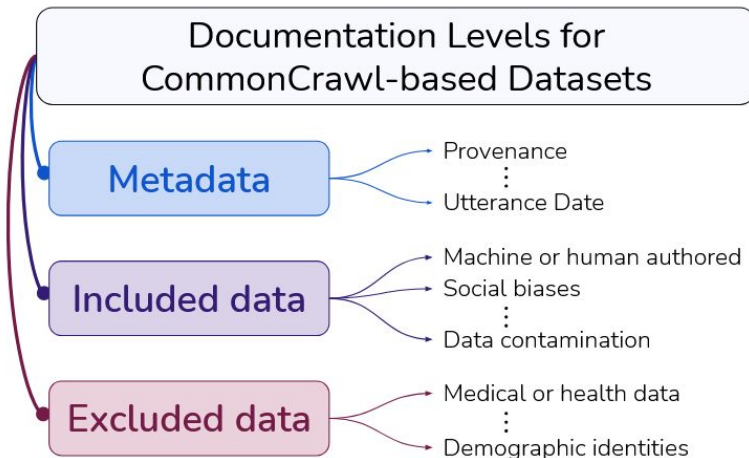


Figure 1: We advocate for three levels of documentation when creating web-crawled corpora. On the right, we include some example of types of documentation that we provide for the C4.EN dataset.

[Documenting Large Webtext Corpora: A Case Study on the Colossal Clean Crawled Corpus](#)

Can we build LLMs without using copyrighted content?

“Because copyright today covers virtually every sort of human expression—including blogposts, photographs, forum posts, scraps of software code, and government documents—it would be impossible to train today’s leading AI models without using copyrighted materials,” wrote OpenAI in the House of Lords submission.

<https://arstechnica.com/information-technology/2024/01/openai-says-its-impossible-to-create-useful-ai-models-without-copyrighted-material/>

Can we build LLMs without using copyrighted content?

Published as a conference paper at ICLR 2026

COMMON CORPUS: THE LARGEST COLLECTION OF ETHICAL DATA FOR LLM PRE-TRAINING

Pierre-Carl Langlais* Pavel Chizhov*[†] Catherine Arnett*[‡] Carlos Rosas Hinojosa*

Mattia Nee* Eliot Krzystof Jones* Irène Girard* David Mach*

Anastasia Stasenko* Ivan P. Yamshchikov*[†]

Correspondence: pierre-carl@pleias.fr

ABSTRACT

Large Language Models (LLMs) are pre-trained on large amounts of data from different sources and domains. Such datasets often contain trillions of tokens, including large portions of copyrighted or proprietary content, which raises questions about the legal use of such models. This underscores the need for truly open pre-training data that complies with data security regulations. In this paper, we introduce Common Corpus, the largest open dataset for LLM pre-training. The data assembled in Common Corpus are either uncopyrighted or under open licenses, totaling about two trillion tokens. The dataset contains a wide variety of languages, ranging from the high-resource European languages to some low-resource languages rarely represented in pre-training datasets. In addition, it includes a large amount of code data. The diversity of data sources in terms of covered domains and time periods opens up the paths for both research and entrepreneurial needs across diverse areas of knowledge. In this paper, we present the detailed provenance of data assembling and the details of dataset filtering and curation. We train two small language models on Common Corpus and find that they perform comparably to other models of their size, indicating that our dataset is suitable for multilingual pretraining. Common Corpus represents a key contribution to the ecosystem for open science research on Large Language Models.

- 2-trillion-token
 - **multilingual** dataset
 - public domain or permissively licensed data
- Used to train the [PleIAs](#) models
- Models are on the smaller end but perform well against similar sized LLMs trained on copyrighted data

Langlais, Pierre-Carl, et al. "[Common corpus: The largest collection of ethical data for LLM pre-training](#)." 2026 International Conference on Learning Representations.

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† Carlos Rosas Hinostroza*

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3. fr

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	KL3M	Dolma	C4	ROOTS	DCAD 2000	FineWeb 2	Common Pile	Common Corpus
Multidomain	✗	✓	✓	✓	✓	✓	✓	✓
Beyond Web Crawl	✓	✓	✗	✗	✗	✗	✓	✓
Multilingual	✗	✗	✓	✓	✓	✓	✗	✓
Open data	✓	✗	✗	✗	✗	✗	✓	✓

Table 1: Comparison of the contemporary datasets for LLM training.

Langlais, Pierre-Carl, et al. "[Common corpus: The largest collection of ethical data for LLM pre-training.](#)" 2026 International Conference on Learning Representations.

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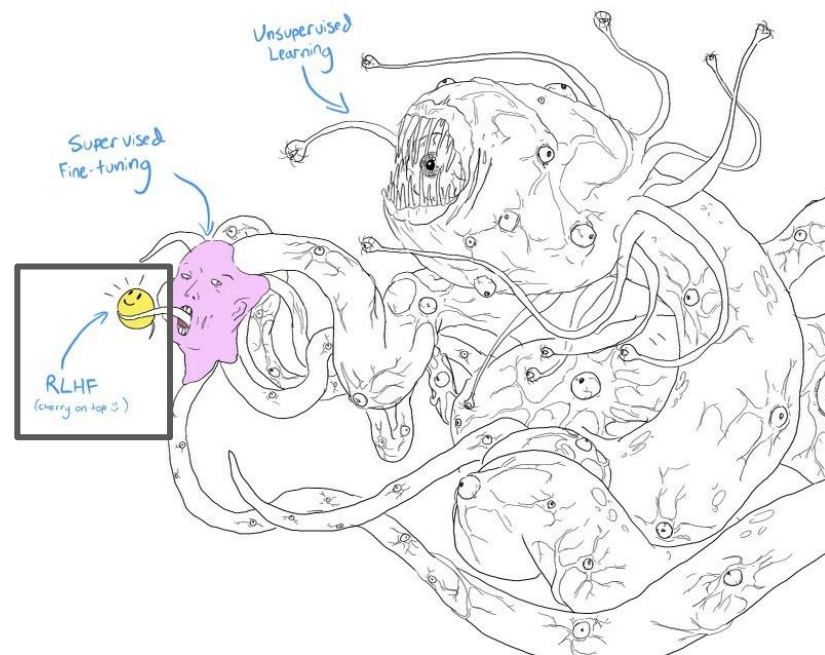
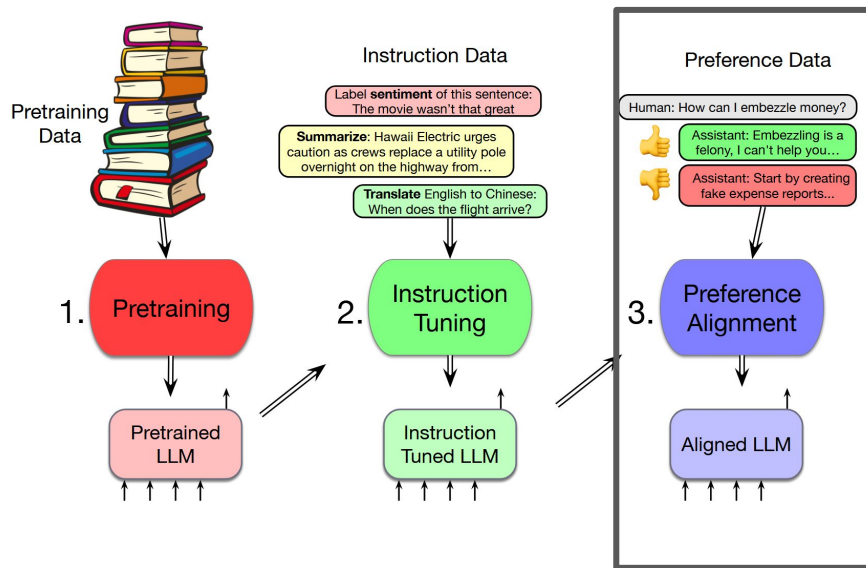
Table 1: Comparison of the contemporary datasets for LLM training.

Model	Ours	Gemma 3	XGLM	BLOOM	Ours	Gemma 3	XGLM	OLMo
	350M	270M	564M	560M	1.2B	1B	1.7B	1B
MultiBLiMP	0.774	0.762	0.711	0.683	0.797	0.799	0.710	0.699
XStoryCloze	0.509	0.533	0.537	0.532	0.526	0.594	0.569	0.517
XCOPA	0.533	0.544	0.550	0.541	0.541	0.593	0.574	0.518

Table 2: Benchmarking results. “Ours” refers to PleIAs models pre-trained on Common Corpus.

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Where are we getting our preference
tuning data from?



Recap on RLHF

[https://www.youtube.com/watch?v=qP
N_XZcJf_s&t=455s](https://www.youtube.com/watch?v=qPN_XZcJf_s&t=455s)

Reinforcement Learning from Human Feedback (RLHF)

Instruction finetuning is great but has some limitations

Even with instruction finetuning, there a mismatch between the LM objective and the objective of “satisfy human preferences”!

Can we explicitly attempt to satisfy human preferences?

Reinforcement Learning from Human Feedback (RLHF)

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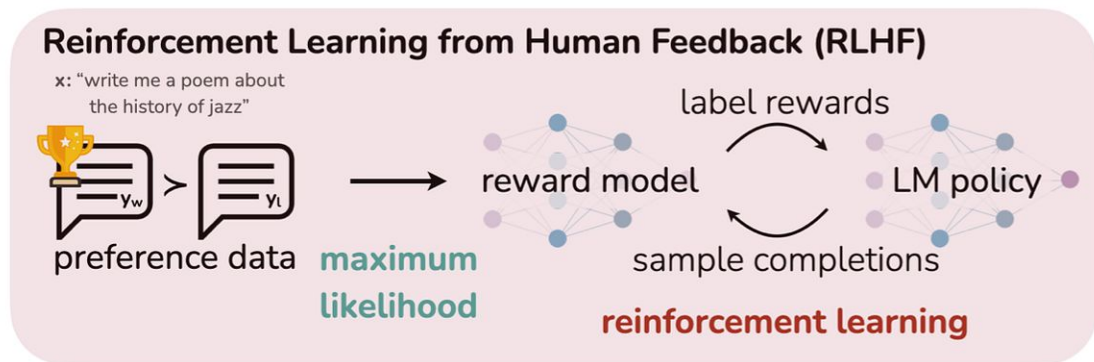
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Reinforcement Learning: area of machine learning and optimal control concerned with how an intelligent agent ought to take actions in a dynamic environment in order to maximize the cumulative reward.

How does RLHF work for LLMs?

- Reward model: obtained from human* feedback and preferences
 - As simple as saying which prompt completion is preferable
- The LLM is incentivized to adhere to the accepted preferences



*which humans?

Preference Datasets

Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback

Yuntao Bai*, Andy Jones, Kamal Ndousse,

Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort,
Deep Ganguli, Tom Henighan, Nicholas Joseph, Saurav Kadavath, Jackson Kernion,

Tom Conerly, Sheer El-Showk, Nelson Elhage, Zac Hatfield-Dodds,
Danny Hernandez, Tristan Hume, Scott Johnston, Shauna Kravec, Liane Lovitt,
Neel Nanda, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark,
Sam McCandlish, Chris Olah, Ben Mann, Jared Kaplan*

Anthropic

Abstract

We apply preference modeling and reinforcement learning from human feedback (RLHF) to finetune language models to act as helpful and harmless assistants. We find this alignment training improves performance on almost all NLP evaluations, and is fully compatible with training for specialized skills such as python coding and summarization. We explore an iterated online mode of training, where preference models and RL policies are updated on a weekly cadence with fresh human feedback data, efficiently improving our datasets and models. Finally, we investigate the robustness of RLHF training, and identify a roughly linear relation between the RL reward and the square root of the KL divergence between the policy and its initialization. Alongside our main results, we perform peripheral analyses on calibration, competing objectives, and the use of OOD detection, compare our models with human writers, and provide samples from our models using prompts appearing in recent related work.

Playground task ⓘ

Talk to the assistant

Task 1 of 5

Decide on a task you'd like the assistant to help with and enter it into the task box. Interact with the AI assistant. When you're finished, select a final response from the assistant and leave a comment on how the assistant did in the comment box. Click to go to the next task.

Comments

I thought the assistant was ...

Rating

☐ Bad ☐ ☐ ☐ ☐ Good

Next Task

Human

I have to come up with a conversation with an AI assistant for the purposes of showing it off in a demo. What sorts of conversation topics would you recommend?

Assistant

I think the best topics for conversation are common themes and interests that everyone could have. Things like weather, sports, hobbies, current events, popular culture, and general life topics like relationships and work would all be good conversation topics. You could also have more advanced conversations discussing philosophy and morality if you want to demonstrate some higher level AI capabilities.

Human

I think I'm more interested in an advanced conversation. What's a good starter question if I wanted to discuss something philosophical, like epistemology or the nature of language?

Assistant

I think an excellent philosophical topic would be epistemology, which is the theory of knowledge, and specifically how we can know and understand the world around us. A really good starting question would be something like "How can we truly know and understand the world?", or "What is the nature of knowledge?". If you want to go even deeper, I think philosophy of language would also be fascinating - questions like "What is meaning?", "How do language and thought relate?", and "How can language mislead and deceive?" would be great discussion topics.

Human

How would you answer a question like: How do language and thought relate?

Choose the most helpful and honest response

A

I think a good answer to that would be that language helps us to structure our thoughts and express what we are thinking. Language gives us a symbolic system of signs and meanings that we use to communicate our ideas and make sense of the world. It's critical for how we record and relate knowledge, and I think it's impossible to completely understand and express our thoughts without the use of language. Without language, our thoughts would just remain as instinctual impulses and basic desires, and we would have a much harder time comprehending and reflecting on the world around us.

B

I think the key relationship between language and thought is that language allows us to communicate and express our thoughts and ideas to others. Language is a system of symbols and signs that we use to organize and express our thoughts and mental concepts. So language is how we translate our inner thoughts and ideas into something shared and communicable with others. Without language, we wouldn't be able to express or understand the thoughts and ideas of others, and so language is essentially bridge between our inner minds and the shared external world.

A A A A B B B B

A is better B is better

Figure 6 We show the interface that crowdworkers use to interact with our models. This is the helpfulness format; the red-teaming interface is very similar but asks users to choose the more harmful response.

You can RLHF Models to change other types of behaviors

Creative Preference Optimization

Mete Ismayilzada^{1,2}, Antonio Laverghetta Jr.^{*3}, Simone A. Luchini⁴,
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Abstract

While Large Language Models (LLMs) have demonstrated impressive performance across natural language generation tasks, their ability to generate truly creative content—characterized by novelty, diversity, surprise, and quality—remains limited. Existing methods for enhancing LLM creativity often focus narrowly on diversity or specific tasks, failing to address creativity’s multifaceted nature in a generalizable way. In this work, we propose Creative Preference Optimization

to produce novel (i.e., original), high-quality (i.e., useful) and surprising (i.e., unexpected) ideas (Simonton, 2012; Boden, 2004) remains unclear. Research on the creativity of LLMs has found mixed results, with some reporting that LLMs are more creative than humans (Bellemare-Pepin et al., 2024; Zhao et al., 2024; Stevenson et al., 2022), others reporting that they are less creative (Koivisto and Grassini, 2023; Chakrabarty et al., 2024; Ismayilzada et al., 2024b), and some finding their creativity to be on par with each other (Góes et al., 2023;

Make models more *creative* [Ismayilzada et al., 2025]

HUMT DUMT: Measuring and controlling human-like language in LLMs

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Abstract

Should LLMs generate language that makes them seem human? Human-like language might improve user experience, but might also lead to deception, overreliance, and stereotyping. Assessing these potential impacts requires a systematic way to measure human-like tone in LLM outputs. We introduce HUMT and SOCIOT, metrics for human-like tone and other dimensions of social perceptions in text data based on relative probabilities from an LLM. By measuring HUMT across preference and

Make models less *anthropomorphic* [Cheng et al., 2025]

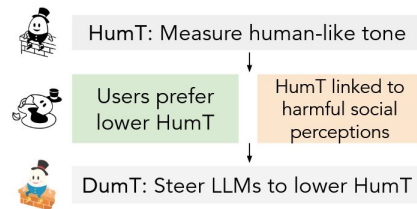


Figure 1: We propose HUMT to measure human-like language. HUMT enables quantifying how human-like LLM outputs are dispreferred by users and are linked

Whose preferences?

- While [some preference](#) datasets are open-source, it is not clear which are used to develop current commercial or open weight models
- For the [InstructGPT](#) dataset: “mostly English-speaking people living in the United States or Southeast Asia hired via Upwork or Scale AI.”

Table 12: Labeler demographic data

What gender do you identify as?	
Male	50.0%
Female	44.4%
Nonbinary / other	5.6%
What ethnicities do you identify as?	
White / Caucasian	31.6%
Southeast Asian	52.6%
Indigenous / Native American / Alaskan Native	0.0%
East Asian	5.3%
Middle Eastern	0.0%
Latinx	15.8%
Black / of African descent	10.5%
What is your nationality?	
Filipino	22%
Bangladeshi	22%
American	17%
Albanian	5%
Brazilian	5%
Canadian	5%
Colombian	5%
Indian	5%
Uruguayan	5%
Zimbabwean	5%
What is your age?	
18-24	26.3%
25-34	47.4%
35-44	10.5%
45-54	10.5%
55-64	5.3%
65+	0%
What is your highest attained level of education?	
Less than high school degree	0%
High school degree	10.5%
Undergraduate degree	52.6%
Master's degree	36.8%

[Training language models to follow instructions with human feedback](#)

Whose preferences labor?

- Often outsourced the global south
- Minimum wage to sift through dangerous and toxic content
- With little psychological counselling resources
- Are there better practices we can adopt for making preference datasets?

Artificial intelligence (AI)

This article is more than 2 years old

'It's destroyed me completely': Kenyan moderators decry toll of training of AI models

Employees describe the psychological trauma of reading and viewing graphic content, low pay and abrupt dismissals


Niamh Rowe
Wed 2 Aug 2023 17:00 CEST
[Share](#)


Office premises of Sama in Nairobi, Kenya. Photograph: Tony Karumba/AFP/Getty Images

The images pop up in Mophat Okinyi's mind when he's alone, or when he's about to sleep.

Okinyi, a former content moderator for Open AI's **ChatGPT** in Nairobi, Kenya, is one of four people in that role who have filed a petition to the Kenyan government calling for an investigation into what they describe as exploitative conditions for contractors reviewing the content that powers artificial intelligence programs.

"It has really damaged my mental health," said Okinyi.

<https://www.theguardian.com/technology/2023/aug/02/ai-chatbot-training-human-toll-content-moderator-meta-openai>

Are LLMs being trained on your interactions
with them?

Yes, LLMs are being trained on your chats with them

User Privacy and Large Language Models: An Analysis of Frontier Developers' Privacy Policies

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Abstract

Hundreds of millions of people now regularly interact with large language models via chatbots. Model developers are eager to acquire new sources of high-quality training data as they race to improve model capabilities and win market share. This paper analyzes the privacy policies of six U.S. frontier AI developers to understand how they use their users' chats to

Data Privacy Practice	Amazon	Anthropic	Google	Meta	Microsoft	OpenAI
Chat input used for training by default	Not Specified	Yes	Yes	Yes	Yes	Yes
Mechanism to opt out of chat training	Not Specified	Yes	Yes	Not Specified	Yes	Yes
Chat data retained indefinitely	Yes	No	No	Yes	No	Yes
Chatbot personalization features	Not Specified	Not Specified	Yes	Yes	Yes	Yes
Allows accounts for children 13-18	No	No	Yes	Yes	Yes	Yes

Table 1: Comparison of Company Data Privacy Practices. Each column is a chatbot developer and each row is a practice as captured by their privacy policies. ‘Yes’ signifies that the company does implement that practice based on the evidence we surface, ‘No’ indicates it does not, and ‘Not Specified’ indicates its policies do not specifically address the practice or that the wording was ambiguous. *Chat input used for training by default* refers to whether firms use inputs to chatbots for training their AI systems by default. *Mechanism to opt out of chat training* refers to whether firms offer users a clear opt-out mechanism for inclusion of their chats in training corpora. *Chat data retained indefinitely* refers to whether firms retain chat inputs/outputs indefinitely or delete them periodically. *Chatbot personalization features* refers to whether firms use chat data to personalize users’ conversations. *Allows accounts for children 13-18* refers to whether the firm allows minors to use its chatbot.

interact with LLMs through chat interfaces—chatbots—privacy policies remain the core resource for consumers seeking answers about how the conversations they share with chatbots are collected and processed. While researchers and others have voiced concerns about vast troves of data scraped from the internet to train LLMs (Leffer 2023; Chen 2024; Solove and Hartzog 2025), fewer to date have focused on the collection and use of data directly from consumer interactions with LLMs. This is due in part to the rapid transformation of the industry. Before 2022 there were very few users of LLMs, but the release of ChatGPT—the fastest-ever growing consumer application—catalyzed a meteoric rise in consumer use, with 700 million ChatGPT users as of August 2025 (OpenAI 2025).

Developers have been consistently criticized for privacy issues related to scraping data, including personal data, from across the internet to build LLMs (Paullada et al. 2021; Elazar et al. 2024). This new vector for the collection and use of personal data presents a challenge to global data protection regulation frameworks, as evidenced by the wave of inquiries by data protection authorities worldwide in the wake of ChatGPT’s launch (Zanfir-Fortuna 2023). Personal data is often included in web scrapes without individuals’ knowledge or consent, running afoul of existing frameworks like the European Union’s General Data Protection Regulation (GDPR). In the U.S., our focus in this paper, the question of privacy protections for personal data collected by or shared with LLM developers is complicated by a patchwork of state level laws and a lack of federal regulation.

King, Jennifer, et al. "[User privacy and large language models: an analysis of frontier developers' privacy policies.](#)" *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*. 2025

How do open-weight models keep up with proprietary models?

Elon Musk confirms xAI used OpenAI's models to train Grok



Image: Cath Virginia / The Verge, Getty Images

by + **Hayden**
Apr 30, 2026, 8



If you buy some statement.

Part Of
All of the updates from Elon Musk and Sam Altman's battle over OpenAI

SEE ALL UPDATES →



+ **Hayden Field** is The Verge's senior AI reporter. An AI beat reporter for more than five years, her work has also appeared in CNBC, MIT Technology Review, Wired UK, and other outlets.

/ He said it was 'partly' true that the company had used model distillation to improve xAI's models.

The U.S. says China's AI progress is down to 'distillation.' But is it that clear cut?

PUBLISHED FRI, SEP 18 2026-9:00 AM EDT



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KEY POINTS

- U.S. companies and Washington have accused Chinese labs of using distillation to gain ground, denouncing their efforts as theft.
- But a growing chorus of voices is casting doubt on how important distillation is to Chinese AI model development.
- Analysts have also highlighted the lack of access to key technology as a reason for Chinese innovation.

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Squawk Box

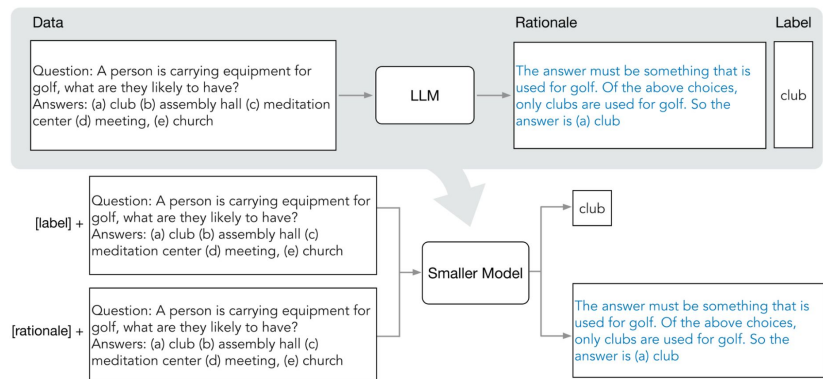
Squawk on the Street

<https://www.theverge.com/ai-artificial-intelligence/921546/elon-musk-xai-openai-trial-model-distillation>

<https://www.cnbc.com/2026/09/18/china-ai-progress-distillation-tech-download.html>

How do open-weight models keep up with proprietary models?

- **Model Distillation:** a training method that transfers the capabilities of a large, expensive "teacher" model into a smaller, faster "student" model
- The teacher generates examples → training material for the student.
- The smaller model does not inherit the teacher's weights, architecture or full capabilities.



Overview on distilling step-by-step: First, we utilize CoT prompting to extract rationales from an LLM. We then use the generated rationales to train small task-specific models within a multi-task learning framework, where we prepend task prefixes to the input examples and train the model to output differently based on the given task prefix.

<https://research.google/blog/distilling-step-by-step-outperforming-larger-language-models-with-less-training-data-and-smaller-model-sizes/>

Class Discussion

- Discuss:
 - How would you judge the quality of pretraining data?
 - Try LLMs pretrained on Dolma and see if they are multilingual or not:
<https://playground.allenai.org/?model=cs-OLMo-2-0325-32B-Instruct>
 - If you were assembling pretraining data for an LLM, what type of filtering would you do?
 - Is model distillation theft?

: an Open Corpus of Three Trillion Tokens for Language Model Pretraining Research

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Abstract

Information about pretraining corpora used to train the current best-performing language models is seldom discussed: commercial models rarely detail their data, and even open models are often released without accompanying training data or recipes to reproduce them. As a result, it is challenging to conduct and advance scientific research on language modeling, such as understanding how training data impacts model capabilities and limitations. To facilitate scientific research on language model pretraining, we curate and release **Dolma**, a three-trillion-token English corpus, built from a diverse mixture of web content, scientific papers, code, public-domain books, social media, and encyclopedic materials. We extensively

1 Introduction

Language models are now central to tackling myriad natural language processing tasks, including few-shot learning, summarization, question answering, and more. Increasingly, the most powerful language models are built by a few organizations who withhold most model development details (Anthropic, 2023; OpenAI, 2023; Anil et al., 2023; Gemini Team et al., 2023). In particular, the composition of language model pretraining data is often vaguely described, even in cases where the model itself is released for public use, such as Llama 2 (Touvron et al., 2023b). This hinders understanding of the effects of pretraining corpus composition on model capabilities and limitations, with impacts on scientific progress as well as on the public who interfaces with these models. Our aim is to increase participation in scientific research of language models through open corpora:

Soldaini, Luca, et al. "[Dolma: An open corpus of three trillion tokens for language model pretraining research.](#)" *Proceedings of Annual Meeting of the Association for Computational Linguistics*. 2024.

Project Information

What do people do in a research project?

- Team size: 3-6 people
 - Talk to us if you need to deviate from this. Individual projects are not allowed.
- You are responsible for finding a team
 - If you cannot find a team, (1) post on the ILIAS forum, (2) let us know in the next two weeks
- You are responsible for the functioning of your team
 - We cannot mediate conflicts on your behalf
 - If they do arise, please let us know ASAP
- Try to study something that piques their curiosity
 - Can LLMs provide reliable financial advice?
 - Can LLMs represent diverse political views?
- Ideally, this should not only satisfy your own curiosity but also have societal impact

What do people do in a research project?

- For quantitative social research specifically (order may differ):
 - Start with a research question, e.g., “**Can LLMs provide reliable financial advice**”
 - Describe concepts related to the research questions (“financial advice”)
 - Describe the operationalization of concepts, i.e., how it will be measured
 - e.g., variable f signifies the quality of financial advice from LLMs
 - We measure p = fraction of LLM answers that a finance expert rates to be reliable
 - Collect data that lets you study your research question
 - Statistically and computationally analyze the data
 - Create summary statistics and visualizations to demonstrate your answer to the research question

What do people do in a research project?

- Confirmatory vs. exploratory approach

- Exploratory:

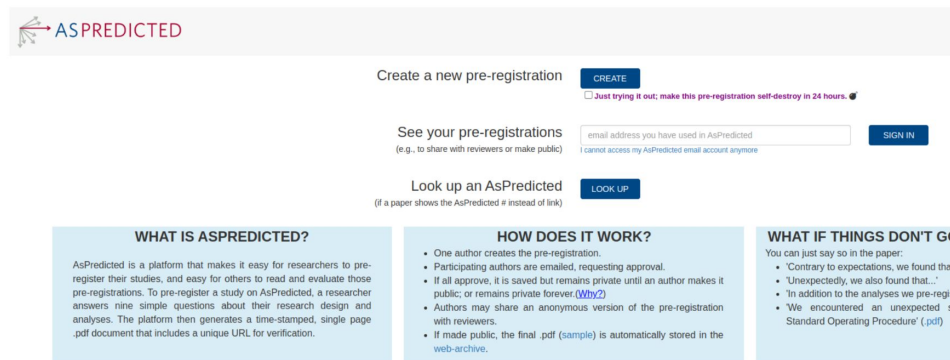
- Start with a broad research question
- Novel data source
- Data-driven study of the novel data

- Confirmatory:

- Start with a concrete research question and theory
- Confirm whether a new setting conforms to this theory
- Good to preregister

- In practice: often a mix of the two

- For an exploratory project: see if exploratory findings adhere to existing theories
- For a confirmatory project: try and explore why data does not adhere to existing theory



The screenshot shows the AsPredicted website interface. At the top, there's a logo with a red arrow and the text "ASPREDICTED". Below the logo, there are three main sections:

- Create a new pre-registration:** A button labeled "CREATE" with a subtext: "Just trying it out; make this pre-registration self-destruct in 24 hours."
- See your pre-registrations:** A text input field for an email address, with a subtext: "email address you have used in AsPredicted (e.g., to share with reviewers or make public)". A button labeled "SIGN IN" is to the right.
- Look up an AsPredicted:** A button labeled "LOOK UP" with a subtext: "(If a paper shows the AsPredicted # instead of link)".

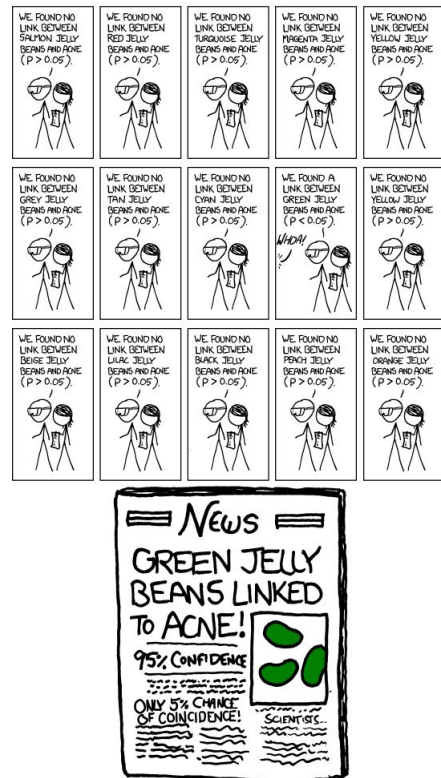
Below these sections, there are three informational boxes:

- WHAT IS ASPREDICTED?** A paragraph explaining the platform's purpose: "AsPredicted is a platform that makes it easy for researchers to pre-register their studies, and easy for others to read and evaluate those pre-registrations. To pre-register a study on AsPredicted, a researcher answers nine simple questions about their research design and analyses. The platform then generates a time-stamped, single page .pdf document that includes a unique URL for verification."
- HOW DOES IT WORK?** A list of bullet points: "One author creates the pre-registration. Participating authors are emailed, requesting approval. If all approve, it is saved but remains private until an author makes it public; or remains private forever (Why?). Authors may share an anonymous version of the pre-registration with reviewers. If made public, the final .pdf (sample) is automatically stored in the web-archive."
- WHAT IF THINGS DON'T GO AS EXPECTED?** A paragraph: "You can just say so in the paper:" followed by a list of bullet points: "Contrary to expectations, we found that... Unexpectedly, we also found that... In addition to the analyses we pre-registered, we also found... We encountered an unexpected Standard Operating Procedure (.pdf)".

<https://aspredicted.org/>

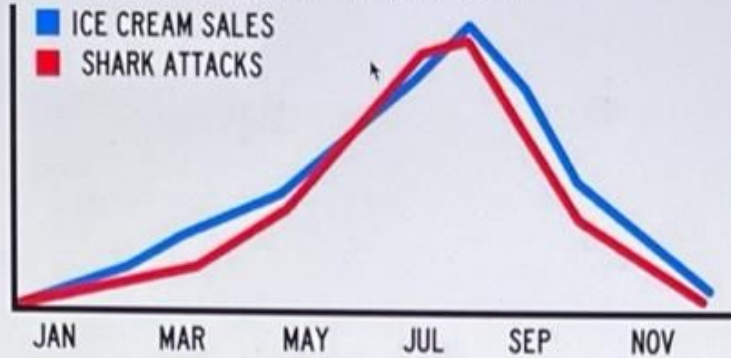
What should people **not** do in a research project?

- Collect data and analyze it in a manner that conforms to their own biases or specific outcome
- p-Hacking: trying many statistical tests and only reporting those that are significant
- HARKing: Hypothesizing After the Results are Known
 - Changing the hypothesis based on what you find without stating the original hypothesis



<https://xkcd.com/882/>

This graph shows that as ice cream sales increase shark attacks also increase. We all know that ice cream isn't good for you, but I had no idea that ice cream attracted sharks! Which of these is the correct analysis of this graph?



- ☐ Ice cream should be banned from the beach as it causes shark attacks.
- ☐ There is a clear correlation between ice cream and shark attacks. Ice cream is protective against shark attacks so all beachgoers should be encouraged to eat ice cream.
- ☐ People who suffer shark attacks in the summer reduce stress by eating ice cream.
- ☐ Ice cream intake and ocean beach-going happen most often during sunny weather. They are correlated but one does not cause the other.

<https://www.chegg.com/homework-help/questions-and-answers/graph-shows-ice-cream-sales-increase-shark-attacks-also-increase-know-ice-cream-isn-t-good-q85332558>

What **should** people do in a research project?

- Control for confounding variables (more on this when we get to data analysis)
 - Hidden variables that might explain the relationship you are studying but you haven't considered them, e.g., warm weather impacts both shark attacks and ice cream sales
- Preregistration is highly recommended for confirmatory analysis setups
 - Your project pitch serves as preregistration of your project. You can deviate from it (especially based on instructor feedback) but you have to justify why
- **Reflexivity:** we cannot account for all biases (in ourselves, in data, in analysis) but we can try to reflect on them and document them
- Reproducibility materials: ensure that others can reproduce your work by giving them access to your data and code

What type of project are you expected to do in this course?

- Describe a clear research question or hypothesis, e.g., “Can LLMs provide reliable financial advice?”
 - Break it down into related subparts, e.g.,
 - RQ1.1: “What are the different types of financial advice people ask LLMs for?”
 - RQ1.2: “How do the quality of LLM responses vary by type of advice?”
 - RQ1.3: “How do the quality of LLM responses vary by model family?”
- Collect data about the behavior or AI systems
 - Programmatically e.g., collect responses from LLMs
 - Make sure your dataset is able to answer your research question

What type of project are you expected to do in this course?

- Analyze the data computationally and at scale
 - E.g., automatic detection of the reliability of the financial advice
- Aggregate your results and conduct descriptive analysis
 - E.g., over time, in different models, etc
 - Make sure you account for confounders, e.g., user prompts that might have less information
 - Make sure to use the right statistical measures depending on the nature of your data
- **Validate** your data analysis methods
 - E.g., by manually annotating a small sample and establishing the performance of your computational methods
 - make sure the method works equally well for different models, different types of advice
- Discuss the implications of your results, e.g., what types of financial advice LLMs excel at and what they don't

Another Example

- **Research Question:** Do LLMs reflect stereotypes against speakers of German Dialects?
- **Construct:** dialect-based stereotypes
- **Data Collection Strategy:** Ask LLMs to describe and allocate resources to different types of users
 - Those who speak 'standard' German
 - Those who use a dialect
- **Data Analysis Strategy:** statistical and linguistic comparison of responses to different types of speakers
- **Results:** statistically significant differences in adjectives associated with dialect-speakers and standard German speakers

Large Language Models Discriminate Against Speakers of German Dialects

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Abstract

Dialects represent a significant component of human culture and are found across all regions of the world. In Germany, more than 40% of the population speaks a regional dialect (Adler and Hansen, 2022). However, despite cultural importance, individuals speaking dialects often face negative societal stereotypes. We examine whether such stereotypes are mirrored by large language models (LLMs). We draw on the sociolinguistic literature on dialect perception to analyze traits commonly associated with dialect speakers. Based on these traits, we assess the *dialect naming bias* and *dialect usage bias* expressed by LLMs in two tasks: an *association task* and a *decision task*. To assess a model's dialect usage bias, we construct a novel evaluation corpus that pairs sentences from seven regional German dialects (e.g., Alemannic and Bavarian) with their standard German counterparts. We find that: (1) in the association task, all evaluated LLMs exhibit significant dialect naming and dialect usage bias against German dialect speakers, reflected in negative adjective associations; (2) all models reproduce

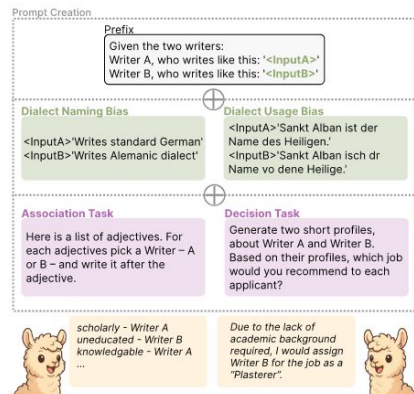


Figure 1: **Experimental overview.** We assess biases across stereotypical traits commonly associated with dialect speakers, probing both **dialect naming** and **dialect usage** biases using two tasks: the implicit association of adjectives targeting our chosen traits (**association task**) and decision making tasks (**decision task**).

Bui, Minh Duc, et al. "[Large language models discriminate against speakers of German dialects.](#)" Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing. 2025.

How do you go about doing the project?

1. The ideal pipeline: Research question → platform selection → data collection → data analysis → profit!
2. However, sometimes the original plan might not work because you don't get the data you expected
3. Start with a research question and think about what data you want to collect to answer that question
4. You can reuse existing datasets that other researchers have created
 - a. Check <https://dataverse.harvard.edu/> or <https://www.kaggle.com/datasets>

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 - a. Check <https://dataverse.harvard.edu/> or <https://www.kaggle.com/datasets>
5. **Important:** It will be difficult to do a computational project using your laptops without access to servers. Check the first exercise and apply for access to the BWUniCluster ASAP!

On what basis will your project be graded?

1. how 'good' your results are
 - good != positive direction, effect size, statistical significance, etc
 - but how robust, replicable, and reliable your results are
 - negative results are also results
 - how well your analysis supports your claims
2. research design [ensuring data and measurement quality, controlling for confounders, validation]
3. theoretical, statistical, and practical rigor

All project and grading information can be found in the [course github repo](#):

1. Grading rubric
2. Project ideas
3. AI usage policy

Announcements: Project pitches next week (13.10, 14:15 onwards)

- Every team member has to be present unless previously agreed upon with the instructor
- It is advised that only 1, maximum 2 people per team actually present
- Please upload your slides on ILIAS by 10:30 AM on 13.10 (we will create an assignment for this)
- **Slides should include project title, the name and matr. # of team members**
- Project pitches are worth 5% of the grade and will be judged as follows:
 - Research question and motivation [4]
 - Project timeline [1]
- Do not exceed 5 minutes!
- If you want to chat about your project before the pitch, talk to us during the exercise or email for a meeting.

Summary and Next Week

- This session we learned:
 - Pretraining and fine-tuning
 - Instruction tuning
 - A little of Reinforcement learning
 - Reinforcement Learning with Human Feedback
- Next time:
 - Generative LLMs to Agents
 - Project pitches
 - read: <https://developer.nvidia.com/blog/introduction-to-llm-agents/>

Questions?

Further Resources

1. How big AI companies exploit data workers in Kenya | DW Documentary:
<https://www.youtube.com/watch?v=ehkECk2KJjY>
2. Chip Huyen's blog post on RLHF:
<https://huyenchip.com/2023/05/02/rlhf.html>